

MOCK TEST: SPORTS AND NUTRITION

Class 12 Physical Education | Chapter 5

MAX MARKS: 50 TIME: 1 HOUR TOPIC: REVISION SET

SECTION A: Very Short Answer (1 Mark Each)

Q1. What is the primary function of carbohydrates in a sports nutrition diet? [1]

Q2. Name one micronutrient essential for oxygen transport in the blood during exercise. [1]

Q3. Define "balanced diet" in the context of sports performance. [1]

Q4. List one food source rich in proteins for muscle repair. [1]

Q5. What is the recommended daily water intake for an active athlete? [1]

SECTION B: Short Answer (3 Marks Each)

Q6. Explain the role of proteins in sports nutrition. Provide one example of a protein-rich food and its benefit. [3]

Q7. Discuss the importance of hydration during sports activities. Mention two signs of dehydration. [3]

Q8. Describe how fats contribute to energy production in endurance sports. Give one example. [3]

Q9. How does pre-competition nutrition affect athletic performance? Provide two tips. [3]

Q10. Outline the key components of a post-exercise meal for recovery. Mention one nutrient and its purpose. [3]

SECTION C: Long Answer (5 Marks Each)

Q11. Elaborate on the macronutrients required for sports performance. Discuss their sources, functions, and recommended intake. [5]

Q12. Analyze the role of micronutrients in preventing sports-related deficiencies. Provide examples of vitamins and minerals. [5]

Q13. Explain the concept of ergogenic aids and supplements in sports nutrition. Discuss benefits and ethical considerations. [5]

ANSWER KEY (Reference Guide)

A1: Provide quick energy for muscles during high intensity.

A2: Iron.

A3: Includes all essential nutrients in proportions for health/performance.

A4: Eggs.

A5: Approx. 3-4 liters (varies with activity/climate).

Section B Highlights:

- **Proteins:** Build muscle tissue. Example: Chicken (amino acids).
- **Hydration:** Manages body temp. Signs: Dry mouth, dizziness.
- **Fats:** Energy for marathons once glycogen is low.
- **Pre-Comp:** Optimize stores. Tip: Complex carbs 2-3 hrs prior.
- **Post-Ex Recovery:** Carbs for glycogen and protein for fiber repair.

Section C Detailed Insights:

Macronutrients: Carbs (60-70% for fuel), Proteins (1.2-2.0g/kg for repair), Fats (20-35% for hormones).

Micronutrients: Vit C (immunity), Calcium (bone strength), Iron (oxygen carrying). Deficiencies lead to anemia or fractures.

Ergogenic Aids: Caffeine/Creatine (Performance boost). Ethics: Natural supplements are allowed; steroids are banned and harmful.